

Numerical TESTs for Partial Differential Equations (PDEs)

September 26, 2023

1 Advection equation

1.1 Governing equation

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\mathbf{u}\phi) = 0$$

Please write down its semi-discretized form based on finite volume method and develop numerical solver based on explicit schemes for both temporal and spatial discretizations.

1.2 Numerical Example

1.2.1 Advection equation on a sine wave test

The accuracy of linear advection equation is first evaluated by using a sine function [1] given as

$$\phi_0(x, y) = \sin 2\pi(x + y)$$

and transported with a constant and uniform advection velocity $(u, v) = (1, 1)$. A square domain $[0, 1]^2$ is partitioned by both triangular and quadrilateral elements with periodic boundary condition imposed in $x - y$ directions. The computation is carried out on gradually refined grids with the time step adjusted by $\text{CFL} = 0.2$ throughout all computations. Please present the L_1 errors and convergence rates at $t = 1.0$.

1.2.2 Solid rotation of a complex profile

To assess the capability of a scheme to resolve discontinuities, we compute solid rotation of a complex profile including a slotted disk, a cone and a smooth hump as shown in [2, 3]. The rotated velocity is given as $\mathbf{u} = (0.5 - y, x - 0.5)$ on a square domain of $[0, 1]^2$ which is partitioned by triangular and quadrilateral elements with 100 vertices along each edge. Please plot contour profiles after one round revolution at time $t = 2\pi$.

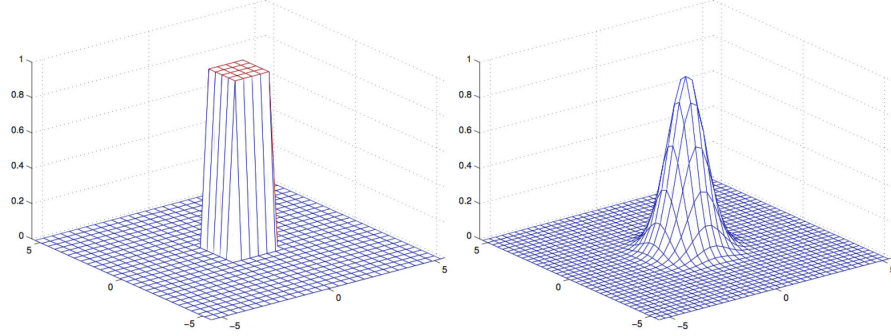


Figure 1: Initial Conditions at time $t = 0$.

2 Diffusion equation

2.1 Governing equation

$$\frac{\partial \phi}{\partial t} - \nabla \cdot \mu \nabla \phi = 0$$

Please write down its semi-discretized form based on finite volume method and develop numerical solver based on explicit schemes for both temporal and spatial discretizations.

2.2 Numerical Example

2.2.1 Diffusion equation with a discontinuity

We begin with the case with a discontinuity where its exact solution for the diffusion equation of $\mu = 1$ is derived in [4] as follow

$$\phi(\mathbf{x}, t) = \frac{1}{4} \left(-\text{Erf} \left(\frac{-1-x}{2\sqrt{t}} \right) + \text{Erf} \left(\frac{1-x}{2\sqrt{t}} \right) \right) \left(-\text{Erf} \left(\frac{-1-y}{2\sqrt{t}} \right) + \text{Erf} \left(\frac{1-y}{2\sqrt{t}} \right) \right).$$

The initial condition of this case is shown in the left-side of Fig. 1. The numerical tests are carried out with Neumann boundary conditions on a square domain of $[-5, 5]^2$ which is divided by triangular and quadrilateral elements with gradually refined resolution. Please measure the solution accuracy by numerical errors and convergence rates at time $t = 0.2$.

2.2.2 Diffusion equation with continuous function

To assess the numerical accuracy, we define a continuous function that satisfies the diffusion equation. The right-side of Fig. 1 shows the initial conditions for this case. A decreasing exponential function is defined as

$$\phi(\mathbf{x}, t) = \frac{1}{1+4t} e^{-\left(\frac{\mu r^2}{1+4t}\right)}$$

where $r = x^2 + y^2$ and $\mu = 1$. A square domain of $[-5, 5]^2$ is partitioned by both triangular and quadrilateral elements with dirichlet boundary condition imposed in $x - y$ directions. The computation is carried out on gradually refined grids with the time step adjusted by its stability condition throughout all computations. Please analyze the accuracy and convergence rate of numerical results at time $t = 0.2$.

3 Advection and diffusion equation

3.1 Governing equation

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\mathbf{u}\phi) - \nabla \cdot \mu \nabla \phi = 0$$

Please write down its semi-discretized form based on finite volume method and develop numerical solver based on explicit schemes for both temporal and spatial discretizations.

3.2 Numerical Example

3.2.1 Advection-diffusion equation by a sinewave function

The accuracy of numerical fluxes is examined by 2D advection-diffusion equation [5] with initial condition $\phi(x, y, 0) = \sin(2\pi(x + y))$ and periodic boundary condition. The exact solution $\phi(x, y, t) = e^{-8\pi^2\mu t} \sin(2\pi(x + y - 2at))$ is transported by a uniform velocity $\mathbf{u} = (a, a)$ with $\mu = 1$ being specified. The test is performed on a square domain of $[0, 1]^2$ which is divided into quadrilateral and triangular grids with varying mesh resolutions. Please measure the solution accuracy by numerical errors and convergence behavior at time $t = 1$.

3.2.2 Advection-diffusion equation with a sharp peak

Next, we consider the time-dependent problem [6] by solving advection-diffusion equation on a square domain of $[-1, 1]^2$ with the velocity field $\mathbf{u} = (-y, x)$ and diffusion coefficient $\varepsilon = 10^{-3}$. The initial and boundary conditions are given by

$$\begin{aligned} \phi(\mathbf{x}, t) &= \frac{1}{4\pi\varepsilon t} e^{-\left(\frac{r^2}{4\varepsilon t}\right)}, \\ r^2 &= (x - \hat{x})^2 + (y - \hat{y})^2, \end{aligned}$$

where $\hat{x}$ and $\hat{y}$ are the time-dependent coordinates of moving peak

$$\begin{aligned}\hat{x}(t) &= x_0 \cos t - y_0 \sin t, \\ \hat{y}(t) &= y_0 \cos t - x_0 \sin t.\end{aligned}$$

We start the numerical experiments from time $t_0 = \pi/2$ with the peak located at the point $(x_0 = 0, y_0 = 0.5)$. As time goes on, the sharp peak is smeared due to diffusion. Please plot the contour profile after one-round rotation and measure the solution accuracy by numerical errors and convergence behavior.

4 Poisson equation

4.1 Governing equation

$$\nabla \cdot \nabla \phi = \omega$$

Please write down its semi-discretized form based on finite volume method and develop numerical solver based on spatial discretization provided in OpenFOAM.

4.2 Numerical Example

The Poisson equation $\Delta \phi = \omega$ is solved on a domain of $[0, 1]^2$ which is divided into quadrilateral and triangular grids. The exact solution for this problem [7] is given by $\phi_{\text{exact}}(x, y) = \cos(\pi x) \cos(\pi y)$ with Dirichlet boundary condition and the forcing term in the form of $\omega(x, y) = -2\pi^2 \cos(\pi x) \cos(\pi y)$. Please compute the solution field ϕ and measure its accuracy by numerical errors and convergence behavior.

5 Navier-Stokes equation

5.1 Governing equation

$$\nabla \cdot \mathbf{u} = 0 \tag{1}$$

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) = -\frac{1}{\rho} \nabla p + \frac{1}{\rho} \nabla \cdot (\mu \nabla \mathbf{u}), \tag{2}$$

Please write down the solution procedure of Navier-Stokes equations based on projection method [8, 9] and PISO method [10, 11] respectively and develop numerical solvers based on finite volume method with discretization schemes provided in OpenFOAM.

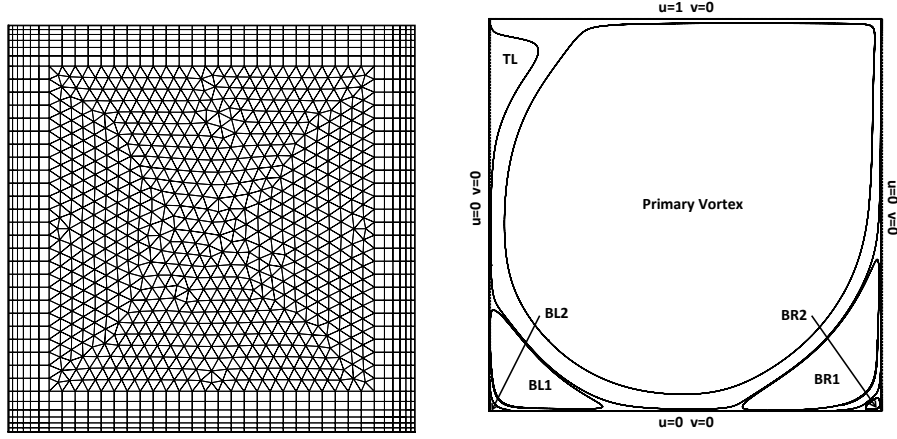


Figure 2: Computational grid and nomenclature of vortices for square lid driven cavity flow.

5.2 Numerical Example

5.2.1 Solution of square lid driven cavity flow

This is the most widely used benchmark test for numerical codes designed to solve viscous incompressible flows [12]. Incompressible flow is enclosed in a square $[0, 1]^2$ domain where nonslip boundary conditions are imposed on the boundary walls. The upper wall ($y = 1$) is moving at a constant speed ($u = 1$, $v = 0$). Numerical tests should be carried out with Reynolds number $1000 \sim 5000$ on hybrid unstructured grids of different resolutions as shown in Fig.2. Please plot the streamline after arriving at the steady state and validate the numerical results by comparing with the reference solutions in [12, 13].

5.2.2 Solution of equilateral triangular lid driven cavity flow

The equilateral triangular cavity is also found in literature[14, 15, 16, 18, 17]. The no-slip conditions are imposed on left and right boundaries, while an unit velocity is prescribed on the top boundary wall. As shown in Fig. 3, a hybrid grid is used again for the purpose of accurately resolving the flow near the boundary. The Reynolds number is defined as Ua/ν , where U is top wall velocity, a is characteristic length, and ν is viscosity. Numerical tests of Reynolds number for 100, 200 and 500 are carried out where a is specified as 1 in all cases. Please plot the streamline after arriving at the steady state and validate the numerical results by comparing with the referred literatures.

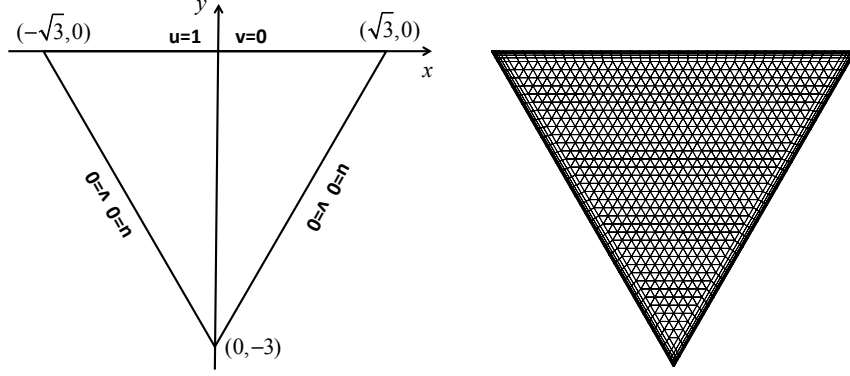


Figure 3: Geometry and computational grid of equilateral triangular lid driven cavity flow.

A Definition of numerical errors and convergence rate

To quantify the numerical errors of results, L_1 , L_2 and L_∞ norms are defined as follows,

$$E(L_1) = \frac{\sum_{i=1}^{N_e} (|\phi_{ni} - \phi_{ei}| |\Omega_i|)}{\sum_{i=1}^{N_e} (|\phi_{ei}| |\Omega_i|)}, \quad (3)$$

$$E(L_2) = \sqrt{\frac{\sum_{i=1}^{N_e} ((\phi_{ni} - \phi_{ei})^2 |\Omega_i|)}{\sum_{i=1}^{N_e} (\phi_{ei}^2 |\Omega_i|)}}, \quad (4)$$

$$E(L_\infty) = \frac{\max |\phi_{ni} - \phi_{ei}|}{\max |\phi_{ei}|}. \quad (5)$$

where ϕ_{ni} and ϕ_{ei} denote numerical and exact solutions respectively. The convergence rate of numerical errors denotes the rate of error reduction with increasing the mesh resolution which are given as

$$N = \frac{\log (\text{Err}_{\Delta 1} / \text{Err}_{\Delta 2})}{\log \left(\sqrt[n]{(\Delta 2 / \Delta 1)} \right)}.$$

The $\text{Err}_{\Delta 1}$ and $\text{Err}_{\Delta 2}$ denote the numerical errors on grids of total number $\Delta 1$ and $\Delta 2$ ($\Delta 2 > \Delta 1$) and n implies the dimensionality, i.e $n = 1$ for one-dimension, $n = 2$ for two-dimensions and $n = 3$ for three-dimensions.

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